

Set theory for SQL Joins

JOINING DATA IN SQL

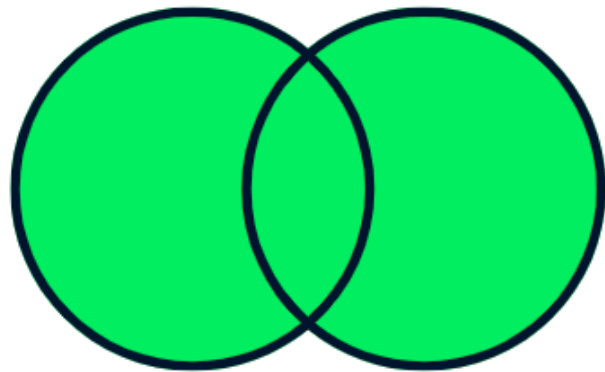


Maham Faisal Khan

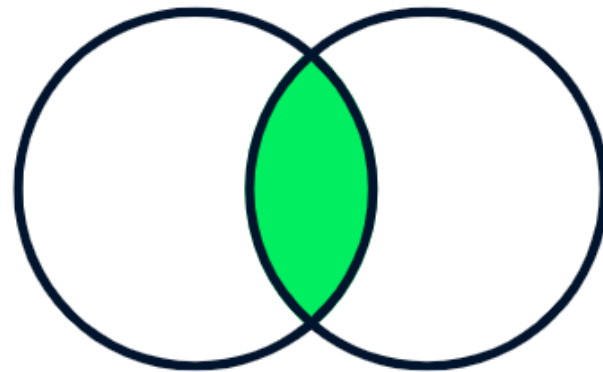
Senior Content Developer, DataCamp

Venn diagrams and set theory

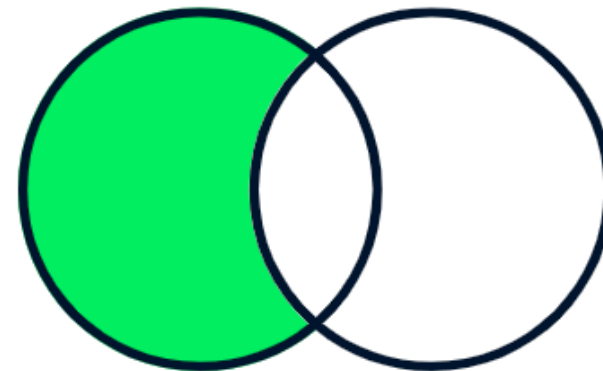
UNION
UNION ALL



INTERSECT

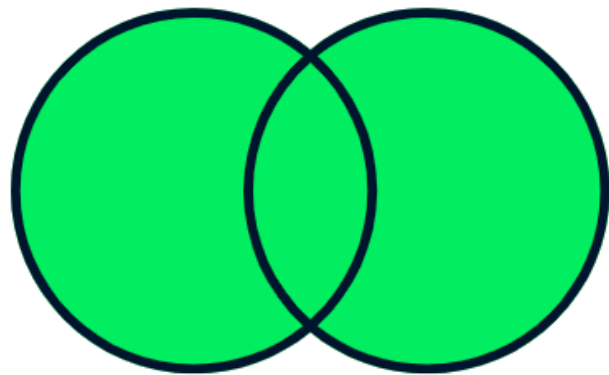


EXCEPT

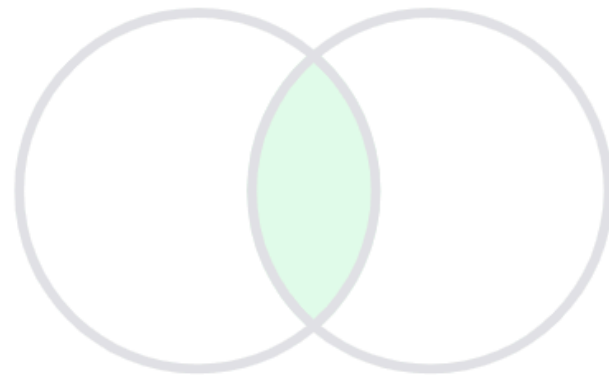


Venn diagrams and set theory

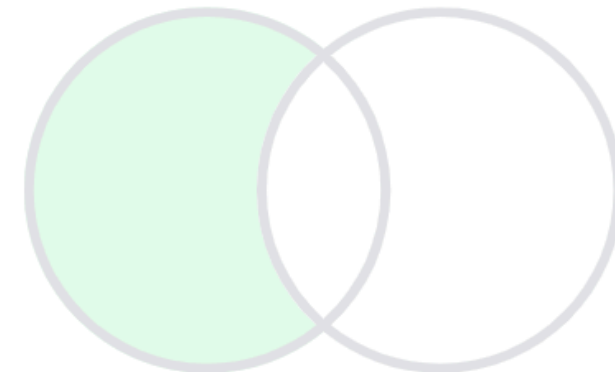
UNION
UNION ALL



INTERSECT

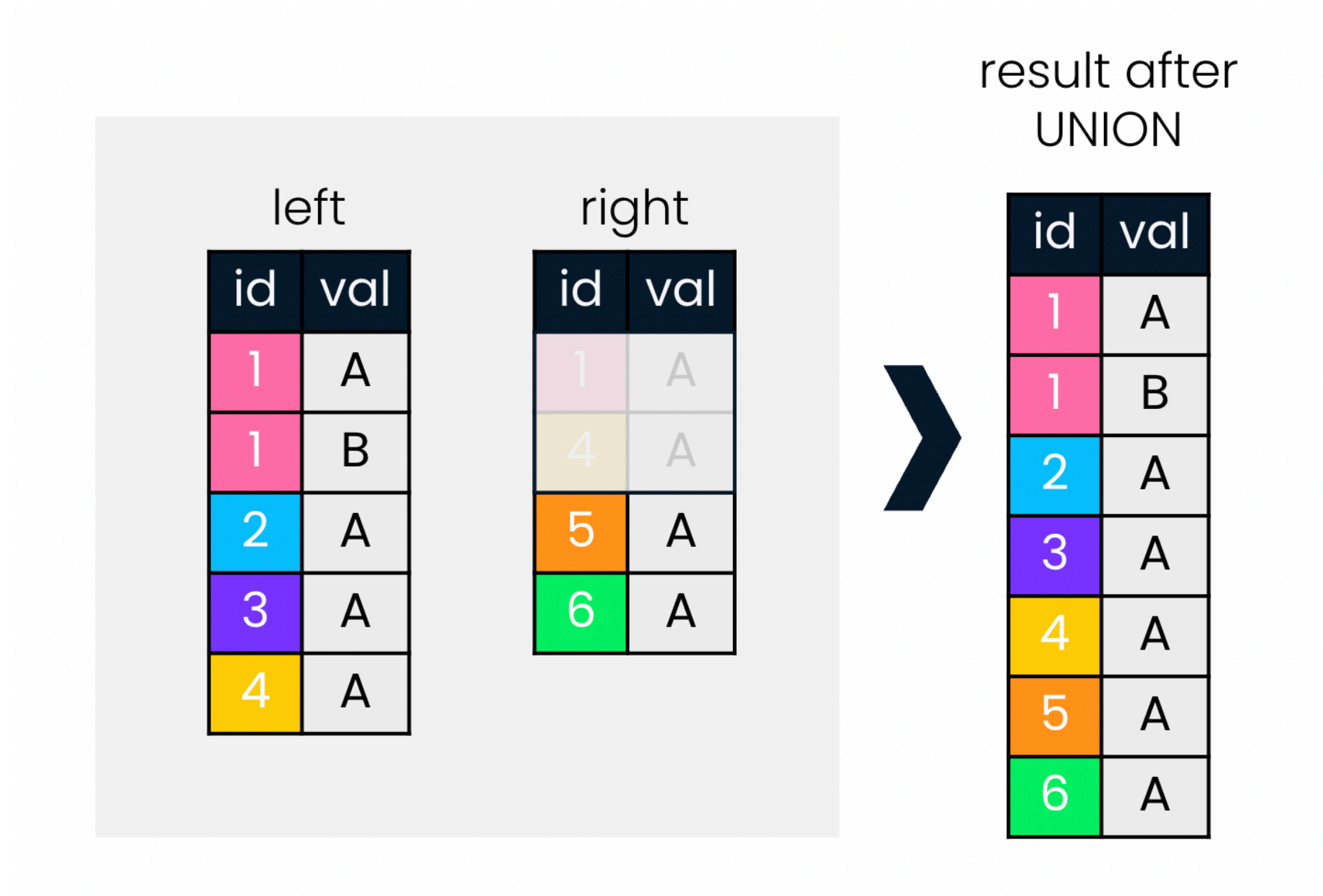


EXCEPT



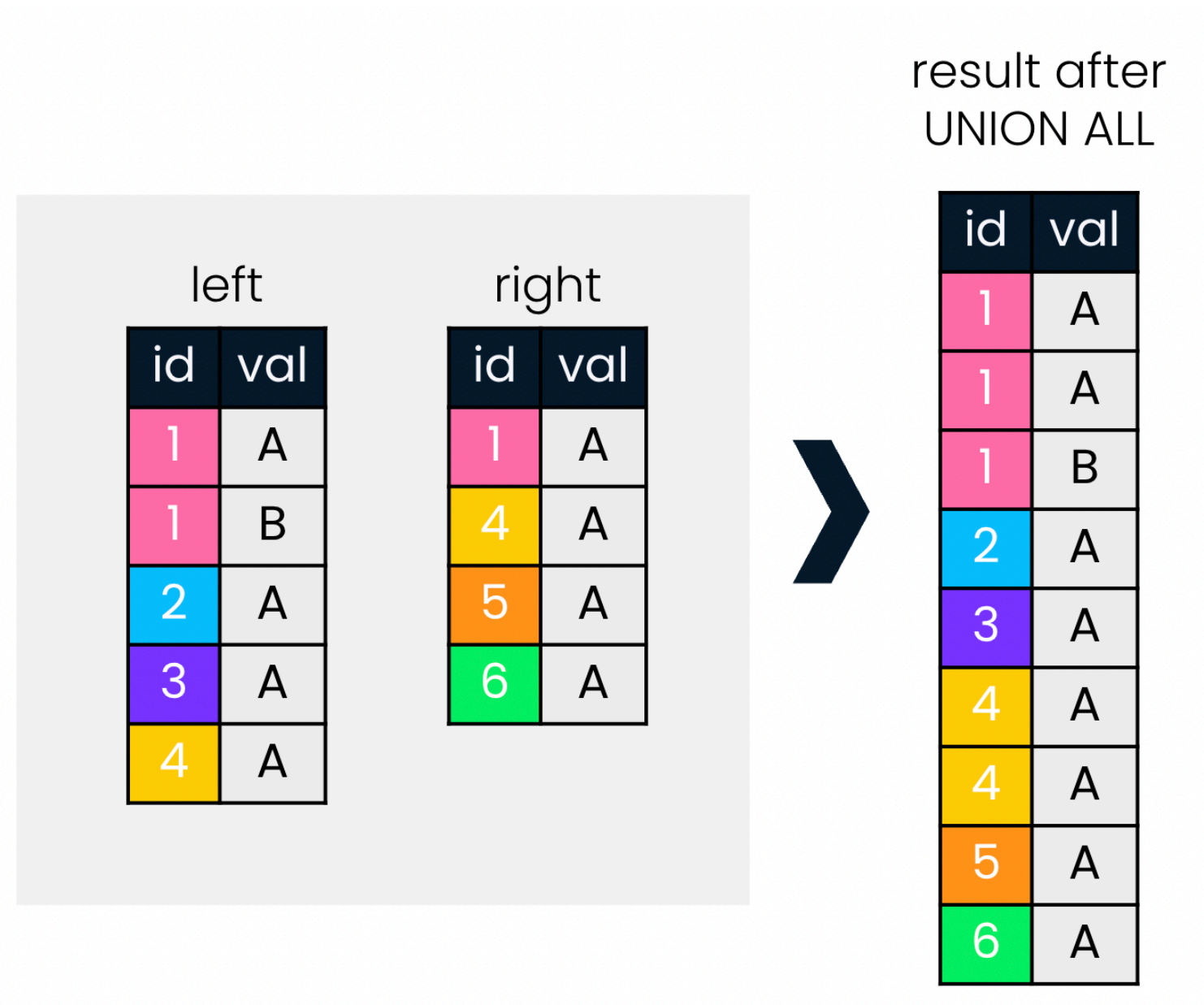
UNION diagram

- `UNION` takes two tables as input, and returns all records from both tables



UNION ALL diagram

- `UNION ALL` takes two tables and returns all records from both tables, including duplicates



UNION and UNION ALL syntax

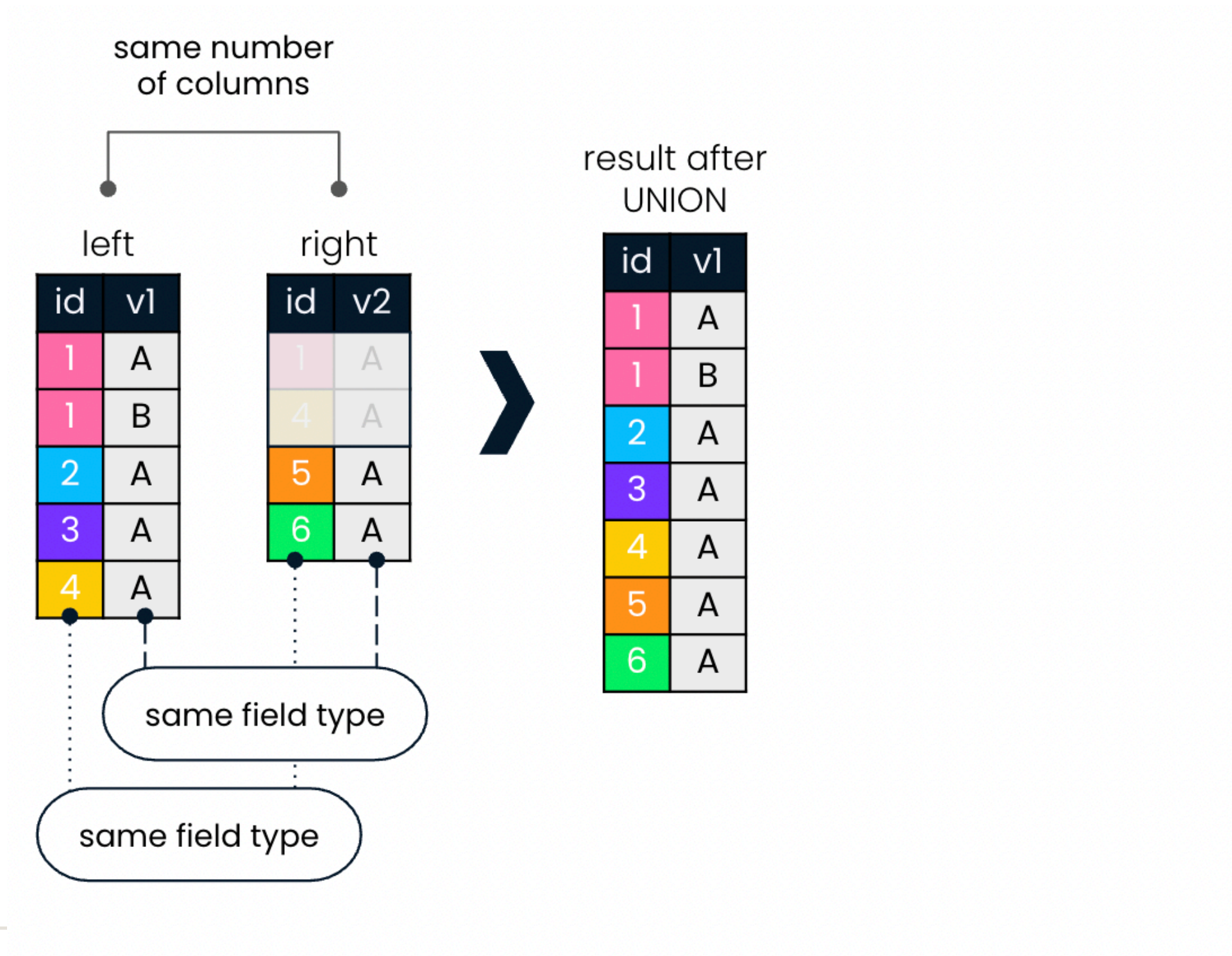
UNION syntax

```
SELECT *  
FROM left_table  
UNION  
SELECT *  
FROM right_table;
```

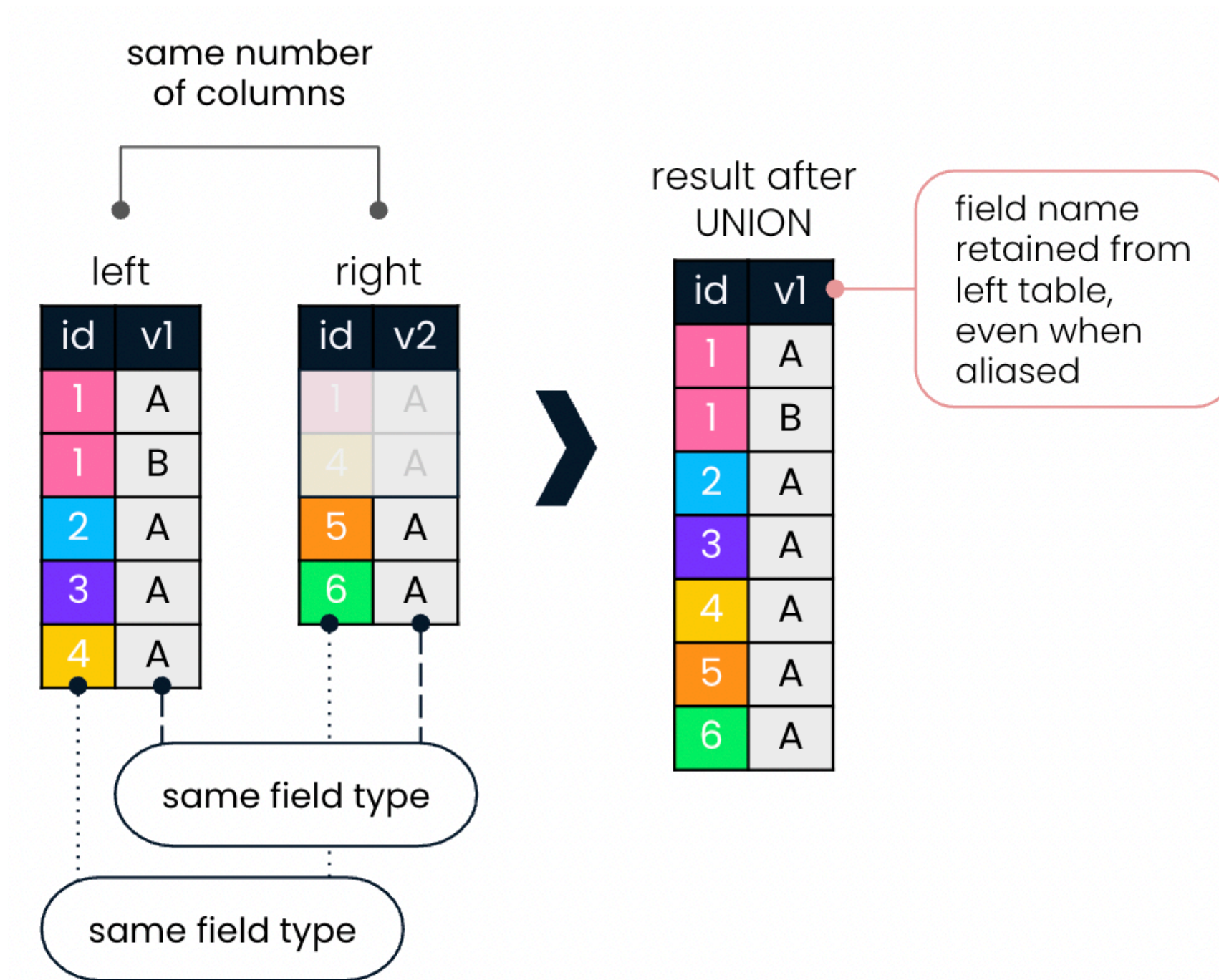
UNION ALL syntax

```
SELECT *  
FROM left_table  
UNION ALL  
SELECT *  
FROM right_table;
```

UNION and UNION ALL syntax



UNION and UNION ALL syntax



To the monarchs table

```
SELECT *  
FROM monarchs;
```

country	continent	monarch
-----	-----	-----
Brunei	Asia	Hassanal Bolkiah
Oman	Asia	Haitham bin Tarik
Norway	Europe	Harald V
United Kingdom	Europe	Charles III

Prime ministers, meet the monarchs

```
SELECT monarch AS leader, country
FROM monarchs
UNION
SELECT prime_minister, country
FROM prime_ministers
ORDER BY country, leader
LIMIT 10;
```

After the UNION

leader	country
Anthony Albanese	Australia
Hassanal Bolkiah	Brunei
Mostafa Madbouly	Egypt
Narendra Modi	India
Christopher Luxon	New Zealand
Harald V	Norway
Jonas Gahr Støre	Norway
Haitham bin Tarik	Oman
Shehbaz Sharif	Pakistan
Luís Montenegro	Portugal

UNION ALL with the leaders

```
SELECT monarch AS leader, country
FROM monarchs
UNION ALL
SELECT prime_minister, country
FROM prime_ministers
ORDER BY leader, country
LIMIT 10;
```

UNION ALL result

leader	country
-----	-----
Anthony Albanese	Australia
Hassanal Bolkiah	Brunei
Hassanal Bolkiah	Brunei
Mostafa Madbouly	Egypt
Narendra Modi	India
Christopher Luxon	New Zealand
Harald V	Norway
Jonas Gahr Støre	Norway
Haitham bin Tarik	Oman
Haitham bin Tarik	Oman

Let's practice!

JOINING DATA IN SQL

At the INTERSECT

JOINING DATA IN SQL

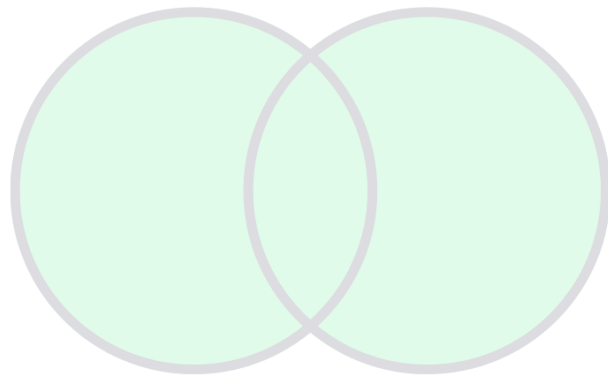


Maham Faisal Khan

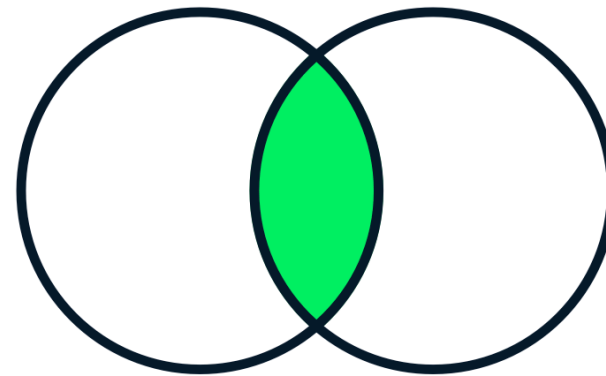
Senior Content Developer, DataCamp

INTERSECT Venn diagram

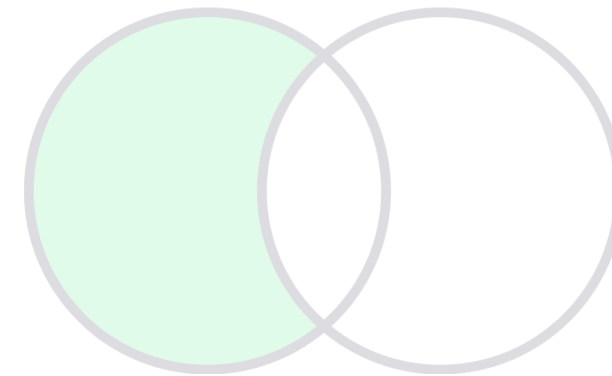
UNION



INTERSECT



EXCEPT



INTERSECT diagram

left_table

id	val
1	N1
1	N1
3	L3
4	L4

right_table

id	val
1	N1
4	R2
5	R3
6	R4



result after
INTERSECT

id	val
1	N1

INTERSECT syntax

```
SELECT id, val  
FROM left_table  
INTERSECT  
SELECT id, val  
FROM right_table;
```

INTERSECT vs. INNER JOIN on two columns

INTERSECT syntax

```
SELECT *  
FROM left_table  
INTERSECT  
SELECT *  
FROM right_table;
```

left_table

id	val
1	N1
1	N1
3	L3
4	L4

right_table

id	val
1	N1
4	R2
5	R3
6	R4



result after
INTERSECT

id	val
1	N1

INNER JOIN syntax

```
SELECT l.id, l.val  
FROM left_table AS l  
INNER JOIN right_table AS r  
  ON l.id = r.id  
     AND l.val = r.val;
```

left_table

id	val
1	N1
1	N1
3	L3
4	L4

right_table

id	val
1	N1
4	R2
5	R3
6	R4



result after
INNER JOIN

id	val
1	N1
1	N1

Countries with prime ministers and presidents

```
SELECT country as intersect_country
FROM prime_ministers
INTERSECT
SELECT country
FROM presidents;
```

```
| intersect_country |
|-----|
| Egypt            |
| Portugal         |
| Pakistan         |
| India            |
```

INTERSECT on two fields

```
SELECT country, prime_minister AS leader
FROM prime_ministers
INTERSECT
SELECT country, president
FROM presidents;
```

----- -----
country leader
----- -----
----- -----

Countries with prime ministers and monarchs

```
SELECT country, prime_minister AS leader
FROM prime_ministers
INTERSECT
SELECT country, monarch
FROM monarchs;
```

```
|-----|-----|
| country | leader   |
|-----|-----|
| Brunei  | Hassanal Bolkiah |
| Oman    | Haitham bin Tarik |
|-----|-----|
```


Let's practice!

JOINING DATA IN SQL

EXCEPT

JOINING DATA IN SQL

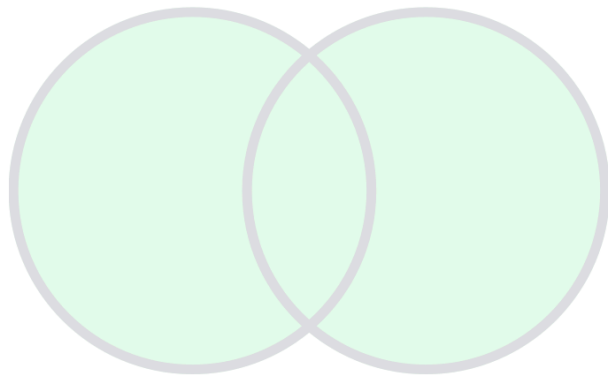


Maham Faisal Khan

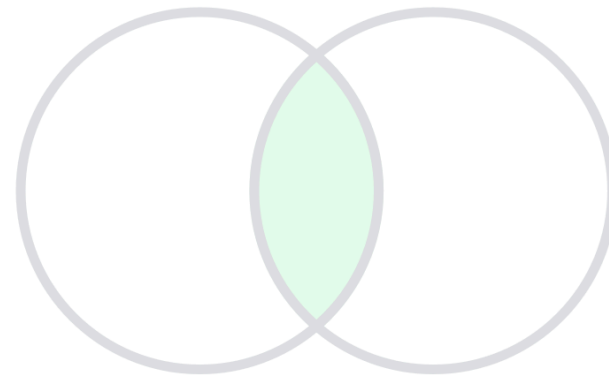
Senior Content Developer, DataCamp

EXCEPT Venn diagram

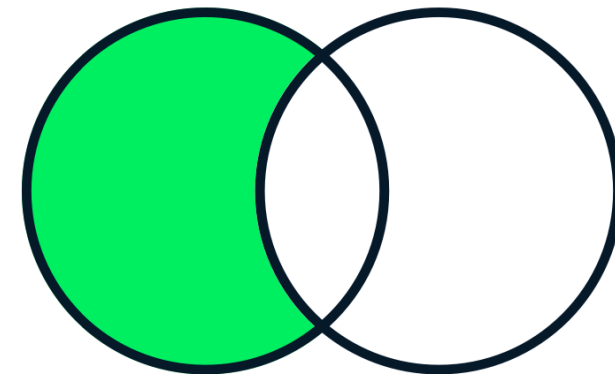
UNION



INTERSECT



EXCEPT



EXCEPT diagram

left_table

id	val
1	N1
1	N1
3	L3
4	L4

right_table

id	val
1	N1
4	R2
5	R3
6	R4



result after
EXCEPT

id	val
3	L3
4	L4

EXCEPT syntax

```
SELECT monarch, country
FROM monarchs
EXCEPT
SELECT prime_minister, country
FROM prime_ministers;
```

----- -----
monarch country
----- -----
Harald V Norway
Charles III United Kingdom
Willem-Alexander Netherlands
----- -----

Let's practice!

JOINING DATA IN SQL

